Lecture 5 (*February 26*)

Reading: Section 2.7
Viewing: Lecture 5

A **permutation matrix** is a square matrix which has a single 1 on each row and column, and 0 everywhere else. For example, the following is a 4×4 permutation matrix:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{40}$$

For all $i \in \{1, \dots, n\}$, look at the single 1 on the i–th row, and let $\sigma(i) \in \{1, \dots, n\}$ be the column where that 1 lies. By the very definition of a permutation matrix, the numbers $\sigma(1), \dots, \sigma(n)$ are all different, hence they determine a permutation of the numbers $1, \dots, n$. For example, the permutation corresponding to the matrix (40) is $(2, 1, 4, 3)$. This is why permutation matrices are called this way, because they are in one-to-one correspondence with usual permutations. Therefore:

$$\text{the number of } n \times n \text{ permutation matrices is } n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1) \cdot n \tag{41}$$

The particular matrix P_{ij} we encountered in Lecture 2 (when discussing the matrix meaning of the various steps in Gaussian elimination) corresponds to the permutation of the numbers $1, \dots, n$ that switches i and j, but keeps all other numbers unchanged.

Suppose you do Gaussian elimination on a square matrix A, always taking care that the pivots are in the right place <u>before</u> doing elimination. For example, we may have:

$$A = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 1 & 2 \\ 1 & 2 & -1 \end{bmatrix}$$

Since the pivot on the first row is to the right of the pivots on the other two rows, let us remedy the situation by exchanging the roles of the first and third rows. As we have seen in Lecture 2, this is achieved by multiplying A on the left with the permutation matrix P_{13}:

$$P_{13}A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

Then we do elimination, by subtracting the first row from the second one:

$$E_{21}^{(-1)} P_{13} A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \\ 0 & 0 & 3 \end{bmatrix} =: U$$

The right-hand side is now in row echelon form, i.e. upper triangular. We may move any product of E's to the right-hand side by multiplying the equation above with their inverses, thus obtaining:

$$P_{13}A = E_{21}^{(1)} U$$

Since $E_{21}^{(1)}$ is a lower triangular matrix and P_{13} is a permutation matrix, this is a particular example of the following general formula, which holds for any non-singular matrix A:

$$\boxed{PA = LU} \tag{42}$$

where P is a permutation matrix, L is lower triangular and U is upper triangular. As we have seen in the example above, P is obtained from A by first rearranging its rows in such a way that the pivots are in the standard order (i.e. ordered left-to-right as we go from top-to-bottom). Then L and U are obtained by the usual Gaussian elimination we described in the previous lecture.

Remark. *Look up to where the word "before" is underlined. If we replace this word by "after", and do Gaussian elimination by first applying elimination matrices and then applying row exchanges, then the factorization we get is $A = LPU$, where P is a permutation matrix. This is a variant of the $PA = LU$ factorization described above.*

Another feature of matrices is the operation of taking **transposes**. Specifically, if A is an $m \times n$ matrix, its transpose A^T is an $n \times m$ matrix obtained by switching the roles of rows and columns:

$$\text{if } A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad \text{then} \quad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

So the first row of A becomes the first column of A^T etc. In terms of entries, the ij entry of A is the ji entry of A^T, for all i and j. Transposes follow certain rules:

$$\boxed{(A+B)^T = A^T + B^T} \qquad \boxed{(AB)^T = B^T A^T} \tag{43}$$

$$\boxed{(A^T)^T = A} \qquad \boxed{(A^{-1})^T = (A^T)^{-1}} \tag{44}$$

(the last property requires A to be square and non-singular, or we couldn't talk about its inverse).

Transposes are related to a lot of things, for example, the dot product of vectors. Compare:

$$\text{the dot product} \qquad \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = a_1 b_1 + \cdots + a_n b_n$$

$$\text{the matrix multiplication} \qquad \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}^T \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 b_1 + \cdots + a_n b_n \end{bmatrix}$$

In other words, given two vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ written in column form, the only entry of the 1×1 matrix $\boldsymbol{v}^T \boldsymbol{w}$ is the same number as the dot product $\boldsymbol{v} \cdot \boldsymbol{w}$.

Another instance of transposes comes about when studying **symmetric** matrices, namely square matrices which are equal to their own transposes:

$$\boxed{S \text{ is called symmetric if } S = S^T} \tag{45}$$

For example, the following matrix is symmetric:

$$\begin{bmatrix} 2 & 5 & -1 \\ 5 & 4 & 8 \\ -1 & 8 & 0 \end{bmatrix}$$

because it is unchanged upon reflecting it across its diagonal. All diagonal matrices (37) are symmetric. More generally, here's a general recipe to obtain symmetric matrices:

$$\boxed{\text{for any matrix } A, \text{ the matrix } S = A^T A \text{ is symmetric}} \tag{46}$$

which holds even if the matrix A is not square. To prove this, note that (using (43) and (44)):

$$S^T = (A^T A)^T = A^T (A^T)^T = A^T A = S$$

Finally, let us note an important reason why symmetric matrices are important from a computational viewpoint. The LU factorization of a symmetric matrix takes a particularly nice form:

$$\boxed{S = LDL^T} \tag{47}$$

where L is a lower triangular matrix with 1's on the diagonal, and D is a diagonal matrix. So in other words, symmetric matrices are precisely those whose LDU factorization has the property that $U = L^T$. You can actually prove that (47) holds for any non-singular symmetric matrix S which admits a LDU factorization, just by using the uniqueness of the LDU factorization. Indeed:

$$S = LDU \quad \overset{(43)}{\Longrightarrow} \quad S^T = U^T D^T L^T$$

It is easy to see that U^T is lower triangular with 1's on the diagonal, L^T is upper triangular with 1's on the diagonal, and $D^T = D$ is diagonal. Since $S = S^T$, the uniqueness of the LDU factorization of $S = S^T$ implies that $U^T = L$ and $L^T = U$ (these two conditions are actually equivalent).